

Maximum Likelihood Estimation for a Gaussian Distribution

COM3031 Probabilistic Machine Learning

1 Problem Setup

Let $\mathcal{D} = \{y_1, y_2, \dots, y_N\}$ be a set of independent and identically distributed samples drawn from a univariate Gaussian distribution $y_i \sim \mathcal{N}(\mu, \sigma^2)$, where μ and σ^2 are unknown parameters.

The objective of Maximum Likelihood Estimation (MLE) is to find parameter values that maximise the likelihood function.

2 Likelihood and Log-Likelihood

The Gaussian probability density function is

$$p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right).$$

Since the samples are i.i.d., the likelihood function is

$$\mathcal{L}(\mu, \sigma^2) = \prod_{i=1}^N p(y_i \mid \mu, \sigma^2).$$

Taking the logarithm gives the log-likelihood:

$$\ell(\mu, \sigma^2) = \sum_{i=1}^N \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y_i - \mu)^2}{2\sigma^2} \right].$$

3 Optimisation via Partial Derivatives

To find the maximum likelihood estimates, we take **partial derivatives** of the log-likelihood with respect to each parameter and set them equal to zero.

3.1 Partial Derivative with Respect to the Mean

$$\begin{aligned} \frac{\partial \ell(\mu, \sigma^2)}{\partial \mu} &= \frac{\partial}{\partial \mu} \sum_{i=1}^N \left[-\frac{(y_i - \mu)^2}{2\sigma^2} \right] \\ &= \sum_{i=1}^N \frac{y_i - \mu}{\sigma^2}. \end{aligned}$$

Stationary Condition

$$\frac{\partial \ell}{\partial \mu} = 0 \implies \sum_{i=1}^N (y_i - \mu) = 0.$$

Solving:

$$N\mu = \sum_{i=1}^N y_i.$$

MLE of the Mean

$$\hat{\mu}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N y_i$$

3.2 Partial Derivative with Respect to the Variance

$$\begin{aligned} \frac{\partial \ell(\mu, \sigma^2)}{\partial \sigma^2} &= \frac{\partial}{\partial \sigma^2} \sum_{i=1}^N \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y_i - \mu)^2}{2\sigma^2} \right] \\ &= -\frac{N}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^N (y_i - \mu)^2. \end{aligned}$$

Stationary Condition

$$\begin{aligned} \frac{\partial \ell}{\partial \sigma^2} &= 0 \\ -\frac{N}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^N (y_i - \mu)^2 &= 0. \end{aligned}$$

Multiplying both sides by $2(\sigma^2)^2$:

$$-N\sigma^2 + \sum_{i=1}^N (y_i - \mu)^2 = 0.$$

MLE of the Variance

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{\mu})^2$$

4 Final Estimates

$$\begin{aligned} \hat{\mu}_{\text{MLE}} &= \frac{1}{N} \sum_{i=1}^N y_i, \\ \hat{\sigma}_{\text{MLE}}^2 &= \frac{1}{N} \sum_{i=1}^N (y_i - \hat{\mu})^2. \end{aligned}$$

5 Interpretation

- MLE chooses parameters that make the observed data most probable.
- For a Gaussian: Best mean = centre of the data and Best variance = average squared deviation
- This links directly to least squares and later to MAP estimation when priors are added